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MILITARY SPECIFICATION

FUEL OIL RECLAIMED

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Fuel Oil, Reclaimed (Stock Number NSN 9140-01-068-6903) which is produced as a product of Navy reclamation operations (product use is described in 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. Unless otherwise specified, the following standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS
FEDERAL
FED-STD-313 - Material Safety Data Sheets
FED-STD-791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commanding Officer, Navy Petroleum Office, 5010 Duke Street, Cameron Station, Alexandria, VA 22304-6180, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9140

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MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290 - Packaging of Petroleum and Related Products

Unless otherwise indicated, copies of Federal and Military specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.

2.2 Non-Government Publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 86 - Distillation of Petroleum Products, Method for (DOD) adopted)
D 88 - Saybolt Viscosity
D 93 - Flash Point by Pensky-Martens Closed Tester, Test Method For (DOD adopted)
D 97 - Pour Point of Petroleum Oils, Test Method For (DOD adopted)
D 129 - Sulfur in Petroleum Products (General Bomb Method), Test Method For (DOD adopted)
D 130 - Detection of Copper Corrosion from Petroleum Products by the Copper Strip Tarnish Test, Method For (DOD adopted)
D 287 - API Gravity of Crude Petroleum and Petroleum Products (Hydrometer Method), Test Method For (DOD adopted)
D 445 - Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity), Test Method For (DOD adopted)
D 473 - Sediment in Crude and Fuel Oils by Extraction
D 482 - Ash from Petroleum Products, Test Method For (DOD adopted)
D 524 - Ramsbottom Carbon Residue of Petroleum Products, Test Method For (DOD adopted)
D 611 - Aniline Point and Mixed Aniline Point of Petroleum Products and Hydrocarbon Solvents, Test Method For (DOD adopted)
D 613 - Ignition Quality of Diesel Fuels by the Cetane Method, Test Method For
D 665 - Rust-Preventing Characteristics of Steam-Turbine Oil in the Presence of Water, Test Method For (DOD adopted)
D 874 - Sulfated Ash from Lubricating Oils and Additives
D 974 - Neutralization Number by Color-Indicator Titration, Test Method For (DOD adopted)

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D 1298 - Density, Relative Density (Specific Gravity), or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method, Test Method For (DOD adopted)
D 1401 - Emulsion Characteristics of Petroleum Oils and Synthetic Fluids, Test Method For
D 1500 - ASTM Color of Petroleum Products (ASTM Color Scale), Test Method For (DOD adopted)
D 1552 - Sulfur in Petroleum Products (High-Temperature Method), Test Method For (DOD adopted)
D 1796 - Water and Sediment in Fuel Oils by Centrifuge Method (Laboratory Procedure)
D 2274 - Oxidation Stability of Distillate Fuel Oil (Accelerated Method), Test Method For
D 2276 - Particulate Contaminant in Aviation Turbine Fuels, Test Method For (DOD adopted)
D 2500 - Cloud Point of Petroleum Oils, Test Method For (DOD adopted)
D 2622 - Sulfur in Petroleum Products (X-Ray Spectrographic Method), Test Method For (DOD adopted)
D 2709 - Water and Sediment in Distillate Fuels by Centrifuge
D 2880 - Gas Turbine Fuel Oils, Standard Specification For
D 3701 - Hydrogen Content of Aviation Turbine Fuels by Low Resolution Nuclear Magnetic Resonance Spectrometry
D 4057 - Standard Method of Sampling Petroleum and Petroleum Products (manual) (DOD adopted)
D 4176 - Free Water and Particulate Contamination in Distillate Fuels (Clear and Bright Pass/Fail Procedures), Test Method For
D 4177 - Standard Method of Sampling Petroleum and Petroleum Products (automatic) (DOD adopted)
D 4294 - Sulfur in Petroleum Products by Non-Dispersive X-Ray Fluorescence Spectrometry, Test Method For
E 29 - Recommended Practice for Indicating Which Places of Figures Are to Be Considered Significant in Specified Limiting Values (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 General. Requirements contained herein are not subject to corrections for tolerance of test methods. If multiple determinations are made by the inspecting laboratory, average results will be used except for those test methods where repeatability data are given. In those cases, the average value derived from the individual results that agree within the repeatability limits given may be used at the discretion of the inspection authority, provided an indication is given of the total number of results obtained and the number falling outside the repeatability limits. For purposes of determining conformance with each requirement, an observed value or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off procedures given in ASTM E 29.

3.2 Material. Fuel Oil, Reclaimed shall consist of a mixture of distillates and residual fuel and may contain used lubricants or other used oil products. The product must be in conformance with 40 CFR 266 for On-Specification Used Oil Fuel to meet the requirements of Fuel Oil, Reclaimed. A product that is subject to regulation as a hazardous waste or that is mixed with a hazardous waste does not meet the requirements of this specification.

3.2.1 Additives. The additives listed herein may be present singularly or in combination as a result of different fuel sources.

3.2.1.1 Stabilizer Additives. The storage stability additive conforming the DOD-A-24682 (SH) may have been blended into the original fuel product.

3.2.1.2 Antioxidants. The following active inhibitors may be present separately or in combination in the fuel typically in total concentration not in excess of 8.4 pounds of inhibitor (not including weight of solvent) per 1,000 barrels of fuel (9.1 grams/100 gallons (U.S.)) in order to prevent the formation of gum:

a. 2,6 - ditertiary butyl-4-methylphenol
b. 2,4 - dimethyl-6-tertiary butylphenol
c. 2,6 - ditertiary butylphenol
d. 75 percent minimum 2,6-ditertiary butylphenol 25 percent maximum tertiary and tritertiary butylphenols

3.2.1.3 Metal Deactivator. A metal deactivator, N,N - disalicyclidene-1, 2 propanediamine may be present in the fuel typically in an amount not to exceed 2 pounds of active ingredient per 1,000 barrels of fuel (2.2 grams/100 gallons (U.S.)).

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3.2.1.4 Ignition Improver. The following additives, to raise the ignition quality of the fuel, may be present in the fuel:

Amyl nitrate (mixed primary nitrates)
Hexyl nitrate (N-hexyl nitrate)
Cyclohexyl nitrate
Octyl nitrate

3.3 Chemical and Physical Requirements. The fuel shall conform to the physical and chemical requirements specified in Table I. The values provided for variability unless otherwise stated.

TABLE I. CHEMICAL AND PHYSICAL REQUIREMENTS.2/4/

Characteristics	Requirements	FED-STD-791 Test Method	ASTM Test Method
Flashpoint, °C (min.)	38		D 93
Explosiveness, % (max.)	50	1151.1	
Water & Sediment, % (max.)	2.0		D 2709
Sediment, % (max.)	0.5		D 473
Density (API Gravity @ 60°C)	(25-40)		D 287
Viscosity at 40°C, Cst	2.0 - 15.0		D 445
Viscosity at 50°C, SU	30 - 90		D 88
Pour Point, °C (max.)	-6/7		D 97
Demulsification at 25°C,3			D 445
Neutrality	Neutral	5101	----
Sulfur content, % (max.)	2.0		D 129
Sulfated Ash, % (max.)	0.15		D 482
Trace metals, ppm (max.)			
Arsenic	5		3040*/3050 EPA SW-846 and proposed method for evaluating solid waste
Cadmium	2		
Chromium	10		
Lead	100		
Total Halogen3/	1000 ppm		ASTM D 808-87

* Recommended only for non-sedimentaceous oils.

2/ In the U.S., limits for sulfur, metals, halogens, PCBs or other constituents shall be as specified or as regulated by EPA, State or local regulations where the fuel is to be burned, whichever is more restrictive. In foreign countries the sulfur limit shall conform to the limit established in the Status of Forces Agreement.

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3/ Fuel Oil, Reclaimed is intended to be managed as an On-Specification Used Oil Fuel under this specification in accordance with regulations for Used Oils provided by 40 CFR 266. Fuel Oil, Reclaimed containing more than 1,000 ppm total halogens is presumed to be a hazardous waste and is not acceptable for use under this specification. This presumption can be rebutted by demonstrating that the oil does not contain significant concentrations of hazardous waste, in which case the oil is acceptable for use under this specification and can be managed as a Used Oil Fuel under 40 CFR 266. Fuel Oil, Reclaimed containing more than 4,000 ppm total halogens that is not a hazardous waste is regulated as an Off-Specification Used Oil Fuel under 40 CFR 266.

4/ Requirements may vary in accordance with changes in Federal, State and local environmental regulations and with specific user equipment requirements.

3.4 Regulatory Requirements. Because Fuel Oil, Reclaimed is a product of a mixture of a variety of oils, the product is subject to regulation by 40 CFR 266, Subpart E. Generators, suppliers and users of Fuel Oil, Reclaimed are also subject to applicable environmental regulations in 40 CFR 266.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility. Unless otherwise specified in the contract or purchase order, the Navy is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the Navy may use its own or any other facilities suitable for the performance of the inspection requirements specified herein.

4.2 Bulk Lot. Bulk lot shall be considered an indefinite quantity of a homogeneous mixture of material offered for acceptance in a single isolated container.

4.3 Homogeneity. The homogeneity of the product will be determined by measuring density (API Gravity) using ASTM Test Method D 287. Lots will be considered homogeneous if the determinations for the upper, middle and lower samples do not vary from the average by more than 0.5 for API Gravity.

4.4 Sampling.

4.4.1 Sampling for Tests. Samples for tests shall be taken in accordance with ASTM D 4057 or 4177. Samples shall be tested in accordance with Table I and 4.5. Upper, middle and lower samples will be taken as described in ASTM D 4057 or 4177. Samples may be composited to a single sample if the lot is homogeneous. If the lot is not homogeneous, specification tests will be performed separately on the upper, middle and lower samples.

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4.5 Inspection. Inspection shall be performed in accordance with method 9601 of FED-STD-791.

4.6 Test Methods. Perform tests in accordance with Table I.

5. PREPARATION FOR DELIVERY

5.1 Packaging, Packing and Marking. Packaging, packing and marking shall be in accordance with MIL-STD-290. The level of packaging, level of packing, type, and size shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended Use. Fuel Oil, Reclaimed can be used as a substitute for ASTM D 396 either directly or as a blend in stationary fuel-burning furnaces for heating buildings, for the generation of steam or other purposes.

6.2 Navy Responsibilities. The Navy is responsible for assuring that the Fuel Oil, Reclaimed meets all the requirements listed in Table I.

6.3 User Responsibilities. The user is responsible for any on-site blending and all operational or equipment modifications necessary to assure that the Fuel Oil, Reclaimed is burned in a safe, efficient and environmentally sound manner as specified in MO-911, "Utilization of Navy Generated Waste Oils as Burner Fuel."

6.4 Subject Term (Keyword) Listing.

Shore Boiler Fuel
F.O.R.
Fuel
Recycle Fuel
Reclaimed Oil

6.5 Changes from Previous Issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 National Stock Number. The following National Stock Number has been assigned to the fuel covered by this specification:

Gallon (COG 1B Item) 9140-01-068-6903

Custodians:
Navy - SA

Review Activities:
Navy - CG, MC, YD1
DLA - PS

Preparing Activity:
Navy - SA
(Project 9140-N107)

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INSTRUCTIONS

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-F-24951A	931117

3. DOCUMENT TITLE

FUEL OIL RECLAIMED

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code*)	
NAVY PETROLEUM OFFICE	(1) Commercial 703-274-7485/6	(2) AUTOVON 284-7485/6
c. ADDRESS (*Include Zip Code*) 5010 DUKE STREET CAMERON STATION T-40 ALEXANDRIA, VA 22304-6180	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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